

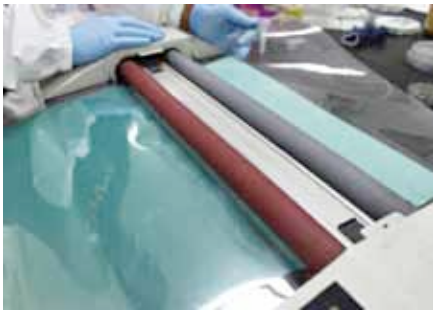


Flexi-displays at CES '11

A hands-on preview of the Samsung OLED flexible display that was showcased at CES last year.
<http://dgit.in/JJS26b>

What the future holds

A sneak peak at what our future will look like in terms of innovative display technology.
<http://dgit.in/Kag9Jy>



Graphene transferred on polyester sheet

after several rounds of exfoliation. Since then, the methods are only getting better and profitable as scientists are continuously trying to move from lab-scale to industrial-scale production. Researchers at Samsung and Sungkyunkwan University, in Korea, have successfully produced a 63 square cm graphene sheet. They grew a graphene layer on copper foil using chemical vapor deposition (CVD) and then passed the copper foil with graphene, attached to an adhesive polymer support, through a roller dipped in copper etchant. What remained was graphene stuck on flexible polymer support.

The size of the sheet was quite a boost. Once the sheets are rolled out, they are pressed and passed again through a final substrate, in this case, polyethylene terephthalate (PET). When treated by doping with nitric acid, these sheets act as transparent, flexible electrodes which can be used instead of indium tin oxide electrodes.

Once the electrodes used in touchscreens become flexible, it will not be too hard for manufacturers to make an entire device flexible. All they would need is a flexible base, like for instance foldable plastic, to mount sensors on. Not being brittle is also one of its key features that is attracting researchers. This would provide consumers with more durable and thus, long lasting products. Phones and tablets will be able to withstand more strain and pressure. Since graphene can be produced using recyclable copper substrates and easily available resources, it would further bring down the cost of final gadgetry. Combined with its ability to consume minimal power, these screens will be nothing shy of being miracu-

lous. All that is left now is for researchers to tap these capabilities to the maximum and bring it into mainstream market as soon as possible.

Current scenario

Companies all over the world have realised the benefits flexible touchscreens offer and are in a race to successfully monetize them. Leading this race at the moment is Atmel. It's been powering touchscreens on popular Galaxy devices like the Galaxy Tab and the Galaxy Note series. The company has recently showcased its curved touchscreen sensors known as XSense (here's a video <http://dgit.in/KRQh7L>). Samsung is hoping to introduce this innovation first in the market. It's already rumoured that Atmel's touch sensors' production quantities will be available in the third quarter of 2012. And it's not hard to imagine Galaxy S4 as a thinner, lighter and yes, a flexible phone.



Nokia kinetic device

Nokia is already in the competition with its Kinetic device, (not a touchscreen, but a flexible display nonetheless).

LG too, since its announcement of electronic paper display (EPD) is looking forward to flexible touchscreen smartphones and tablets. And considering that Apple has always been secretive about their innovations, we won't be surprised if they decide to hop on the wagon soon. Who knows the upcoming iPhone may blow the competition away with a flexible phone?

Applications

Tablets and smartphones are just a part of the big picture. When the structural constraints for using touchscreens are overcome, it may find applications in almost every field possible. Since it is low cost, we can expect the products to be cheaper and pocket friendly. The scope of flexible touchscreen depends on your imagination. It can replace paper in most applications. A calendar which is capable of storing your to-do list, or a newspaper which will update itself everyday keeping up with the latest news so that we don't have to give up our age old habit of reading from 'paper'. We can mount a flexible touchscreen on closets at our home or at shopping malls to scroll through outfits to see which suits us best.

Graphene also has the potential to replace glass in applications like test tubes and sunglasses. For instance, a test tube will be able to display the contents of the solutions inside it and enable us to monitor them. On a broader sense, we can practically use the technology pretty much everywhere.

Also, the coolness quotient cannot be ignored by the manufacturers in the applications area. Imagine an iPhone which rolls into a wrist watch. These fancy items, which we gaped in awe at in Hollywood flicks and Youtube videos, would certainly become an important factor of competition amongst companies trying to gain an upper hand in the market. With each one trying to outdo the other in terms of efficiency and design, it's a great time to be a technology consumer right now.

If flexible touchscreens were not enough, graphene has also found its applications in tapping solar power. You can charge your phone or tablet by taking it outdoors during the day. That solar-charging app which trolled Android users can become a reality. Though a distant concept, it isn't an impossible one.

Once the ice cream has been whipped and sandwich enjoyed, the focus is bound to shift to display hardware and flexing your touchscreens is surely one good way to go about it. 

REASON

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